

OXIEDO: The Architecture That Unlocks Locked Data

The most valuable data in the world cannot be trained on, because no model can account for what it did with it. OR-

383 Controlled Experiments Review the Natural PS 2026 Workshop it learns. Submitted to AAAI 2027

| One Decision in 1986. Three Problems Everyone Funds Separately.

A Single Global Error Signal Makes Fault Localization Mathematically Impossible — Every Downstream Failure Follows From It

NOTHING CAN BE LOCATED

One error signal spreads across every parameter at once.

No unit can assess its own state.

The loss curve collapses and nothing says which node died.

NOTHING IS RETAINED

New tasks overwrite prior representations indiscriminately.

ResNet-18 keeps 47.3% after a sequential task shift.

Every regime change costs what the model knew.

NOTHING IS REPAIRED

Repair requires knowing what is broken.

Corrupted data is learned silently and confidently.

Rollback or full retrain are the only recourse.

One cause: the chain rule multiplies every unit's contribution through every other unit's. Unchanged since 1986.

| Kill a Converged Layer. One Architecture Comes Back.

Bounding Each Node's Local Gradient Chain Produces Attribution as a Physical Byproduct — and Attribution Makes Repair Possible

STANDARD BACKPROPAGATION

One global error signal, entangled across all parameters.

- No per-node visibility.
- Silent collapse on node death.
- No diagnosis, no targeted repair.

Feature layer zeroed at epoch 100:
10.0% — chance, every seed, zero variance, permanent.

ORMAS THREE-SIGNAL TRAINING

Each node reads through a **4,416-parameter** bottleneck — too small to memorise noise.

- Scores its own health every step.
- Fires at **2.5σ** below its own history.
- Diagnoses 7 modes, repairs mid-run.

Same attack: **80.3%** after 85 targeted corrections (**+70.3pp**).

Local stability characterization from Input-to-State Stability (Sontag, 1989) — not a global convergence proof, not yet peer-reviewed.

| The Industry Is Moving From Inference to Training on Private Data

Three Independent Pressures Converge on the Same Missing Layer — None of Them Existed Two Years Ago

REGULATORS SHIFTED TO ARCHITECTURE

The FDA rejects medical models for opacity **regardless of accuracy**.

The EU AI Act attaches liability to systems with no audit trail.

The requirement is no longer performance. It is provenance.

RENTED INFERENCE STOPPED SCALING

Per-call cost compounds linearly with usage.

At institutional volume the only exit is your own infrastructure.

Which means training on your own data — the thing nobody can do safely.

MODELS DEGRADE INVISIBLY

Every regime change, patient cohort and fraud pattern erodes what the model knew.

No detection exists at the training layer.

It surfaces as a miscalculation, not an alert.

The regime is shifting from inference on public data to training on proprietary institutional data.

| Five Sectors. One Question. Counted Up, Not Down.

These Five Are Where the Missing Account Already Costs Somebody Money, in Figures That Can Be Checked

THE QUALIFYING BUYER

Three properties at once, or not a customer:

1. Data that cannot legally leave.
2. An active plan to train on it.
3. Somebody who will ask what the model learned.

The constraint keeps the count small and the contracts large.

WHAT THE GAP COSTS TODAY

AI Training — \$16K–\$24K per recovery from a silent failure.

Medical AI — 12–36 months added to market entry.

Regulated Finance — no component-level audit, no validation. Programme written off.

Data Obligation — \$100K–\$500K per deletion request.

Defense — “we cannot say how” ends the deployment.

Published third-party figures, one per sector. No buyer interviews yet — the primary open risk, stated rather than assumed away.

| One Product. Three Things Arrive With the Model.

Training Is Unchanged — Data, Objective and Resulting Model Are Identical. What Changes Is the Account That Exists Alongside It

01 — THE MODEL

Deploys exactly as it would have

Same architecture, same inference cost, no change to the serving path.

Nothing is traded away for the accountability. That is the whole point.

02 — THE NAMED PARTS

What it learned, per source

Each source's contribution held as a discrete addressable object, bound to the record of the data that justified it.

Removable on instruction, without retraining.

03 — THE RECORD

Every change training made

Which component, at which step, under which diagnosis, by how much, inside what declared limit.

Signed, and diffable against the last approved version.

Seven features of one architecture, not seven products. Nothing is priced per feature. [anonymous.4open.science/r/ormas-EB73](https://www.reddit.com/r/openai/comments/10ormas/anonymous_4open_science_rormas_eb73/) — all 383 runs reproduce with one command.

I Registered a Company to Try to Disprove My Own Architecture

Seven Escalating Tests Over Twelve Months, Each Designed to Fail in Public — All Seven Held

RUNGS 1–4 REMOVE IT

1. **Origination** — cold start, no brand, no ads. Customers arrived, 100% LLM-referred.
2. **Unit economics** — \$0.0043 per 12-asset suite against \$50–\$150 human.
3. **Causal isolation** — system stripped, everything else identical. —91% output.
4. **Collapse shape** — —100% new customers, every channel on the same day.

RUNGS 5–7 PUT IT BACK

5. **Reversibility** — re-injected, nothing else changed. Recovered.
 6. **Magnitude** — +1,300%, overshooting 3.3× on accumulated memory alone.
 7. **Ceiling** — \$4,386 order closed at \$0.00 CAC, no human in the transaction.
- It stops there because I stopped it** — no jurisdiction assigns liability to an agent.

This tests the orchestration layer — prior work, now closed. It is not evidence for ORMAS, which carries its own record. Twelve-month revenue: \$6,691.68 across 79+ orders. Capped at research grade deliberately.

| Publish the Mechanism. License the Accountability.

Defensibility Never Rested on Secrecy — It Rests on Calibration, Regulatory Acceptance, and a Record a Reimplementation Does Not Come With

PUBLISHED — FREE

The architecture and every experiment behind it release in full on publication.

A result nobody can check is worth nothing to a regulator.

Written into the licence, not promised.

LICENSED — REVENUE

Production deployment, on the customer's own infrastructure.

One integration, one contract, nothing priced per feature.

Zero data leaves the perimeter, so margin is software-grade.

WHAT IS ACTUALLY BOUGHT

Not the architecture — that is free from release day.

A calibrated bound, and a counterparty who will stand behind it in a filing.

Calibration is accumulated runs and failure history. A reimplementation arrives with none of it.

| The Regulator Creates the Demand. A Person Has to Close It.

The Requirement Is Written Down Before We Arrive — What Is Missing Is Someone Who Has Survived a Procurement Review

WHY THEY COME TO US

- Regulators now reject models on architecture, not accuracy. The requirement exists before the first meeting.
- The audit trail is produced during training, not reconstructed afterwards.
- Nothing leaves the client's infrastructure — which removes the objection that kills every hosted alternative.

THE ONE GAP

A regulated institution buys from a person it can hold accountable in the room.

No architecture sits in a compliance review and answers for itself.

That is the co-founder — someone who has sold into a fund, a hospital or an insurer and won.

Design partner → single-model deployment → organisation-wide licence. Regulated cycles run 6–18 months.

Everyone Is Competing One Layer Above the Constraint

Orchestration Is Not the Blocker — the Blocker Is That You Cannot Safely Train on the Data

	Per-node attribution	Stability analysis	Autonomous repair	Audit trail at train time	Their metal
Agent frameworks	×	×	×	×	×
Enterprise data platforms	×	×	×	~	~
Frontier model vendors	×	×	×	×	×
Data labelling vendors	×	×	×	×	×
Academic research	~	~	~	×	—
This	✓	✓	✓	✓	✓

The moat is not secrecy — the work publishes on release. It is that getting here ran through a live company broken on its own data, then control theory, then two years of experiments. That order does not compress.

| Everything Technical Is Built. One Thing Is Not.

Three Years Solo Produced the Architecture, the System and the Causal Proof — the Gap Is a Person, Not a Capability

FOUNDER

Eighteen. No university, no lab, no advisor, no external funding.

A self-correcting architecture with a formal stability result, derived at 17.

Programme committee reviewer, **NeurIPS 2026** workshop.

Every experiment ran on one GPU, bought at 15 with money from the first thing he sold.

CO-FOUNDER SOUGHT

One specific profile:

Has sold software inside a fund, a hospital or an insurer.

Understands model risk well enough to win an argument with a compliance officer.

Not an engineer. Not a generalist.

Equal equity, four-year vest.

The honest gap: never having worked inside a team. raadh.me · zenodo.org/records/21730363

| From a Training Layer to the Substrate Regulated AI Runs On

Every Phase Removes One Dependency — Ending With a Stack That Rents Intelligence From No One

PHASE 1 — NOW

Finish the research and carry per-node health monitoring onto attention heads.

Architecture-agnostic by design, so this is a compute problem, not a research one.

First design partners.

PHASE 2 — 12-24 MONTHS

First licensed deployments inside institutions that cannot move their data.

Each accumulates institutional memory that cannot be exported.

Switching cost compounds with tenure.

PHASE 3 — YEARS 3-5

A model pre-trained on the self-correcting architecture from scratch.

The external dependency disappears; cost collapses to hardware.

A position no competitor reaches by adding a feature.

The network learns to stop needing correction. That is the thesis, and it is measured.

Correction frequency decays $4.2 \rightarrow 0.05$ per epoch — the system repairs itself until it no longer needs to.

| \$1.5M Pre-Seed. Sized to the Next Result, Not the Whole Plan.

Eighteen Months, Six Milestones With Acceptance Criteria — the Compute Line Costed Against the Question Actually Open

USE OF FUNDS

- \$650K Co-founder + research engineer
- \$250K 32–64 accelerators, two campaigns
- \$150K One custodian, non-exclusive
- \$100K Delaware, EU rep, filings
- \$200K 18 months of everything else
- \$150K Buffer, unallocated on purpose

WHAT IT BUYS, IN ORDER

1. **A second and third person.** Key-person risk is total today.
2. **One result that is not CIFAR.** Bounded attribution on a transformer backbone, same adversarial suite.
3. **One custodian saying yes.** No regulated data has ever touched this.

The frontier-scale campaign is a seed-round problem. It is not being asked for here.

Before writing a cheque, spend \$50 instead: the remaining programme is falsifiable in under 30 GPU-hours, on one card, run by your own technical partner on your hardware.